Data Science with Python 3

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Chapter Three- Data Representation

3.6.3 Apache Storm

3.6.4 Samza

3.6.5 Flink

3.7 Questions

OBJECTIVES

1. To familiarize with the data representation and types of data

2. To familiarize with the general techniques of handling data

3.1 BIG DATA ANALYTICS

The assortment تشكيلة of any kind of data which is large, highly complex and difficult to manage and handle in the real time scenario with the traditional management techniques becomes the Big Data.

The most extensively technique for the management of data was **Relational DBMS** which was a kind of dataset that fits in all the scenarios for the storage, manipulation and interpretation of data but it failed in case of big data.

كانت التقنية الأكثر شمولاً لإدارة البيانات هي نظام إدارة قواعد البيانات العلائقية، وهو نوع من مجموعات البيانات التي تناسب جميع السيناريوهات الخاصة بتخزين البيانات ومعالجتها وتفسيرها، ولكنها فشلت في حالة البيانات الضخمة.

A groundbreaking study دراسة رائدة in 2013 reported 90% of the entirety of the world‘s data has been created within the previous two years. In the previous two years researchers have composed and handled data which is 9 times the data being processed in the previous 1000 years of the survival. As per the statistical institute, already 2.7 zettabytes of data have been generated and by 2025 it might escalate to a limit beyond belief which ca be 100 zettabytes or even more.

What do we do with all of this data? **How do we make it useful to us?** What are its real-world applications? These questions are the domain of data science.

3.2 DATA SCIENCE - WORKING

In order to handle the query for complete in depth and a sophisticated approach for the exploration of the raw unprocessed data into the real time multiple disciplines with the expertise in machine learning and AI based dimensions, data science gives the solution with the most advanced means and methods.

The scrutiny of the relevant information from the irrelevant raw data and pass on or forward only the most vital data for the enhancing the computing efficiency with the revolution in the fields of engineering, mathematics, statistics, advanced computing and visualizations.

The data science is the basic field of exploration of data where the researchers or the data scientists also rely heavily on artificial intelligence for the creation of simulated models and then predicting the values of the parameters with the algorithms designed for manipulation of data into information. The working in the field of the data science is thereby based on the types of data being explored and manipulation needed e.g., if the simple data in form of tables is available then RDBMS is used and if image data is present then RDBMS fails.

علم البيانات هو المجال الأساسي لاستكشاف البيانات حيث يعتمد الباحثون أو علماء البيانات أيضًا بشكل كبير على الذكاء الاصطناعي لإنشاء نماذج محاكاة ثم التنبؤ بقيم المعلمات باستخدام الخوارزميات المصممة لمعالجة البيانات وتحويلها إلى معلومات. وبالتالي فإن العمل في مجال علم البيانات يعتمد على أنواع البيانات التي يتم استكشافها والتلاعب المطلوب على سبيل المثال، إذا كانت البيانات البسيطة في شكل جداول متاحة، فسيتم استخدام نظام إدارة قواعد البيانات العلائقية وإذا كانت بيانات الصور موجودة، فإن نظام إدارة قواعد البيانات العلائقية يفشل.

3.3 FACETS OF DATA

In data science and big data, the data required for the manipulation and interpretation changes with the change in the types of tools and techniques applied in the algorithm.

The types of the data explored in the field of data science can be categorized as follows:

- **Structured**
- **Unstructured**
- **Natural language**
- **Machine-generated**
- **Streaming**
- **Acoustic data, video, and images**
- **Sensor Data**
- **Graph-based/ Network data**

As in the previous chapter we have explored the data categories structured versus unstructured data. Let's talk about the rest of the representation of data and the explore the characteristics.

The structured data is the main type of data explored in a model and can be expressed as a record with a field of fixed length and dimension. The RDBMS and the Excel data is usually the structured data with known or used defined datatypes with structural information. Bu the unstructured data is data that may not fit any kind of mathematical or simulated model for the study of data due to the context-specific or varying nature of the stored raw data. One example of unstructured data is the email.

3.3.1 Natural Language Data

Another special type of **unstructured data** is the **Natural language data** with the challenging approach to progression the data and manipulate it.

نوع خاص آخر من البيانات غير المنظمة هو بيانات اللغة الطبيعية ذات النهج الصعب لتقدم البيانات والتعامل بها.

The handling of large amount of natural language data and processing in itself necessitates the data scientists to acquire the knowledge of specific data science techniques and linguistics.

إن التعامل مع كمية كبيرة من بيانات اللغة الطبيعية ومعالجتها في حد ذاتها يتطلب من علماء البيانات اكتساب المعرفة بتقنيات علم البيانات واللغويات المحددة.

The natural language processing community has had success in **entity recognition**, **topic recognition**, **summarization**, **text completion**, and **sentiment analysis**, but models trained in one domain don't generalize well to other domains.

Even state-of-the-art techniques aren't able to decipher the meaning of every piece of text. The meaning of the same words can vary when coming from someone upset or joyous.

لقد حقق مجتمع معالجة اللغة الطبيعية نجاحًا في التعرف على الكيانات، والتعرف على الموضوعات، والتلخيص، وإكمال النص، وتحليل المشاعر، ولكن النماذج المدربة في مجال واحد لا يمكن تعميمها بشكل جيد على مجالات أخرى. حتى التقنيات الحديثة غير قادرة على فك رموز معنى كل جزء من النص. يمكن أن يختلف معنى نفس الكلمات عندما تأتي من شخص منزعج أو سعيد.

3.3.2 Machine Generated Data

The machine or the computer has the capability to generate the data based on the requirement of the developer or the application developed.

تتمتع الآلة أو الكمبيوتر بالقدرة على إنشاء البيانات بناءً على متطلبات المطور أو التطبيق الذي تم تطويره.

The data generated by the machine or any robotic process without the intervention of a human is a fast process and also help to forecast certain information for the application, or other machine without human intervention.

البيانات التي تولدها الآلة أو أي عملية روبوتية دون تدخل الإنسان هي عملية سريعة وتساعد أيضًا في التنبؤ بمعلومات معينة للتطبيق أو أي آلة أخرى دون تدخل بشري.

Machine-generated data is becoming a major data resource and will continue to do so. Wikibon has forecast that the market value of the industrial Internet (a term coined by Frost & Sullivan to refer to the integration of complex physical machinery with networked sensors and software) will be approximately $540 billion in 2020. IDC (International Data Corporation) has estimated there will be 26 times more connected things than people in 2020.

وتوقعت شركة ويكيبون أن تبلغ القيمة السوقية للإنترنت الصناعي (وهو مصطلح صاغته شركة فروست آند سوليفان للإشارة إلى دمج الآلات الفيزيائية المعقدة مع أجهزة الاستشعار والبرامج المتصلة بالشبكة) نحو 540 مليار دولار في عام 2020. وقدرت شركة البيانات الدولية (IDC) أن عدد الأشياء المتصلة سوف يزيد بنحو 26 مرة عن عدد الأشخاص في عام 2020.

This network is commonly referred to as the internet of things. The analysis of machine data relies on highly scalable tools, due to its high volume and speed. Examples of machine data are web server logs, call detail records, network event logs, and telemetry.

يشار إلى هذه الشبكة عادةً باسم إنترنت الأشياء. يعتمد تحليل بيانات الآلة على أدوات قابلة للتطوير بدرجة كبيرة، نظرًا لحجمها الكبير وسرعتها. ومن أمثلة بيانات الآلة سجلات خادم الويب، وسجلات تفاصيل المكالمات، وسجلات أحداث الشبكة، والقياس عن بعد.

3.3.3 Streaming Data

The **streaming data** when being administered for the information can take nay form and format which is an interesting property.

يمكن للبيانات المتدفقة عند إدارتها للمعلومات أن تتخذ أي شكل وتنسيق، وهو ما يعد خاصية مثيرة للاهتمام.

This type of data only gets loaded into the server or the data warehouse when any relevant activity occurs or there's an interpretable change in the parameters. The data is not accessed in batch mode. This type of data is not actually a different category but the system for the processing of such varying formats needs to be adaptive to the minutest variations in the information to be handled.

لا يتم تحميل هذا النوع من البيانات إلى الخادم أو مستودع البيانات إلا عند حدوث أي نشاط ذي صلة أو حدوث تغيير قابل للتفسير في المعلمات. ولا يتم الوصول إلى البيانات في وضع الدفعات. لا يشكل هذا النوع من البيانات فئة مختلفة في الواقع، ولكن النظام المستخدم في معالجة مثل هذه التنسيقات المتنوعة يحتاج إلى التكيف مع أدق الاختلافات في المعلومات التي يتعين التعامل معها.

Examples are the **'What's trending' on Twitter**, live sporting or **music events**, and the stock market.

3.3.4 Acoustic, video and image data

The world of animation and digitization has developed multimedia datasets with **audio**, **video** and **images**.

These types of datsets can be stored, handled and interpreted with the Object-Oriented Databases. The databases include the class inheritance and interface in a pre-specified format for handling the complexity of the data and finding its application in **Digital libraries**, video-on demand, **news-on demand**, musical database, etc.

The other type of information in the multimedia datasets can be represented and reproduced as sensory experiences - touch, sense and hear which are typically leading to storing a huge amount of data handled by digitizing.

Furthermore, data compression for images and sounds can exploit limits on human senses performed by throwing away information not needed for good-quality experience which is performed by compression.

علاوة على ذلك، يمكن لضغط البيانات للصور والأصوات استغلال القيود المفروضة على الحواس البشرية من خلال التخلص من المعلومات غير اللازمة للحصول على تجربة ذات جودة جيدة والتي يتم إجراؤها عن طريق الضغط.

There are certain **limitations** when you deal with such a large amount of data are described below:

1. Range is limited and might lead to misinformation

• only certain pitches and loudness can be heard لا يمكن سماع سوى درجات معينة من الصوت ومستوى الصوت

• only certain kinds of light are visible, and there must be enough / not too much light. يمكن رؤية أنواع معينة فقط من الضوء، ويجب أن يكون هناك ما يكفي من الضوء وليس الكثير منه.

2. Discrimination due to the descriptive features of the data التمييز بسبب السمات الوصفية للبيانات

• pitches, loudness, colors, intensities can't be distinguished unless they are

different enough (color1, color2) لا يمكن التمييز بين النغمات، ودرجة الصوت، والألوان، والشدة إلا إذا كانت مختلفة بدرجة كافية (اللون 1، اللون 2)

3. Coding the information of the sensory data البيانات الحسية into digital world.

• nervous systems "encode" experience, e.g., rods and cones in the eye

"تشفر الأنظمة العصبية الخبرات، على سبيل المثال، القضبان والمخاريط في العين"

3.3.4.1 REPRESENTATION OF IMAGE DATA

The **image data** is expanding vastly and due to enhancement in the cameras the encoding is needed for computation and manipulation of image data. The techniques are vector array-based encoding and bit map-based encoding.

تتوسع بيانات الصور بشكل كبير ونتيجة للتحسينات التي طرأت على الكاميرات، أصبح التشفير ضروريًا لحساب ومعالجة بيانات الصور.

Vector graphics encode the image sequences as a series of **lines** or **curves**. The process is expensive in term of image computation but smoothly rescales.

Bit map encode the image as an **array of pixels**. The encoding process is cost effective in terms of computation but scales inefficiently leading to loss of image data.

The Basic idea of image data is the array of receptors where each receptor مستقبل records a pixel by "counting" the number of photons that strike it during exposure.

الفكرة الأساسية لبيانات الصورة هي مجموعة المستقبلات حيث يقوم كل مستقبل بتسجيل بكسل عن طريق "عد" عدد الفوتونات التي تصطدم به أثناء التعرض.

The Red, green, blue recorded separately at each point on image produced by group of three receptors where each receptor is behind a color filter.

يتم تسجيل اللون الأحمر والأخضر والأزرق بشكل منفصل في كل نقطة في الصورة التي تنتجها مجموعة من ثلاثة مستقبلات حيث يكون كل مستقبل خلف مرشح الألوان.

Representation of Acoustic data البيانات الصوتية:

In recent years, the analysis of acoustic characteristics of speech and sound has been one of the areas that data mining has found its way through.

في السنوات الأخيرة، أصبح تحليل الخصائص الصوتية للكلام والصوت أحد المجالات التي وجد فيها data mining طريقه.

The present research study is also related to this topic which aims to detect the gender of the speaker by using the acoustic feature of his voice.

وتتعلق الدراسة البحثية الحالية أيضًا بهذا الموضوع والتي تهدف إلى الكشف عن جنس المتكلم من خلال استخدام السمة الصوتية لصوته.

When an instrument is played or a voice speaks, periodic (many times per second) changes occur in air pressure, which we interpret as acoustics.

عندما يتم العزف على آلة موسيقية أو يتحدث صوت، تحدث تغيرات دورية (عدة مرات في الثانية) في ضغط الهواء، وهو ما نفسرها على أنها صوتيات.

For the representation of the acoustic data compression is needed. Codecs (compression/decompression) implement various compression/decompression techniques which are either lossy or lossless.

لتمثيل البيانات الصوتية، يلزم الضغط. تنفذ برامج الترميز (الضغط/فك الضغط) تقنيات ضغط/فك ضغط مختلفة إما مع فقدان البيانات أو بدون فقدانها.

The lossy compression of the acoustic data may lead to loss of certain information as it is non-repetitive: MPEG (like JPEG) a family of perceptually-based techniques are all lossy techniques.

قد يؤدي الضغط مع فقدان البيانات الصوتية إلى فقدان معلومات معينة لأنها غير متكررة: MPEG (مثل JPEG) هي عائلة من التقنيات القائمة على الإدراك، وكلها تقنيات مع فقدان البيانات.

While the other type of techniques is where the information of the acoustic data is preserved in which WMA Lossless, ALAC, MPEG-4 ALS methods are applied.

في حين أن النوع الآخر من التقنيات هو حيث يتم حفظ معلومات البيانات الصوتية والتي يتم فيها تطبيق طرق WMA Lossless و ALAC و MPEG-4 ALS.

The encoding of data in form of the sounds heard or the visuals captured are highly challenging for the data scientists to process the information.

يعد تشفير البيانات في شكل أصوات مسموعة أو صور مرئية ملتقطة تحديًا كبيرًا لعلماء البيانات لمعالجة المعلومات.

Some tasks performed by the human brain have been of great difficulty to the computers for the recognition of the object in the images.

لقد شكلت بعض المهام التي يقوم بها الدماغ البشري صعوبة كبيرة بالنسبة لأجهزة الكمبيوتر في التعرف على الأشياء في الصور.

MLBAM (Major League Baseball Advanced Media) announced in 2014 that they'll increase video capture to approximately 7 TB per game for the purpose of live, in-game analytics. The higher resolution cameras and acquiring sensors have the capability to capture the motion of ball and the player in a real time scenario e.g., the path taken by a defender relative to two baselines.

أعلنت MLBAM (Major League Baseball Advanced Media) في عام 2014 أنها ستزيد من التقاط الفيديو إلى حوالي 7 تيرابايت لكل مباراة لغرض التحليلات المباشرة أثناء اللعبة. تتمتع الكاميرات وأجهزة الاستشعار ذات الدقة الأعلى بالقدرة على التقاط حركة الكرة واللاعب في سيناريو في الوقت الفعلي، على سبيل المثال، المسار الذي يسلكه المدافع بالنسبة لخطي الأساس.

Recently a company called DeepMind succeeded at creating an algorithm that's capable of learning how to play video games. This algorithm takes the video screen as input and learns to interpret everything via a complex process of deep learning. It's a remarkable feat that prompted Google to buy the company for their own Artificial Intelligence (AI) development plans. The learning algorithm takes in data as it's produced by the computer game; it's streaming data.

في الأونة الأخيرة، نجحت شركة تدعى DeepMind في إنشاء خوارزمية قادرة على تعلم كيفية لعب ألعاب الفيديو. تأخذ هذه الخوارزمية شاشة الفيديو كمدخل وتتعلم تفسير كل شيء من خلال عملية معقدة من التعلم العميق. إنه إنجاز رائع دفع جوجل إلى شراء الشركة لخطط تطوير الذكاء الاصطناعي الخاصة بها. تأخذ خوارزمية التعلم البيانات أثناء إنتاجها بواسطة لعبة الكمبيوتر؛ إنها بيانات متدفقة.

3.3.5 Sensor Data

Any input acquired from the physical environment which is detected and responded from the devices as an output is the collaborative approach of the sensor data.

أي مدخلات يتم الحصول عليها من البيئة المادية والتي يتم اكتشافها والاستجابة لها من قبل الأجهزة كمخرجات هي النهج التعاوني لبيانات المستشعر.

The extracted data from the sensor devices act as an output for a real time system or as an input to another system for the performance of any activity.

تعمل البيانات المستخرجة من أجهزة الاستشعار كمخرج لنظام الوقت الحقيقي أو كمدخل لنظام آخر لأداء أي نشاط.

These sensors can be used to perceive any type of physical element with the approach to detect the events or changes in the environment. A sensor is always used with other electronics, as simple as a lamp or as complex as a computer. Advanced chip technology makes it possible to integrate all the required functions at low cost, in a small volume and with low energy consumption.

يمكن استخدام هذه المستشعرات لإدراك أي نوع من العناصر المادية من خلال اكتشاف الأحداث أو التغييرات في البيئة. يتم استخدام المستشعر دائمًا مع إلكترونيات أخرى، بسيطة مثل المصباح أو معقدة مثل الكمبيوتر. تتيح تقنية الشريحة المتقدمة دمج جميع الوظائف المطلوبة بتكلفة منخفضة وحجم صغير واستهلاك منخفض للطاقة.

The number of sensors around us is increasing rapidly. Estimates vary, but many expect that by 2030 more than 500 billion sensors will be connected to each other via the Internet of Things (IoT).

The exponential growth of the IoT based systems leads to ever demanding rise in the input sensor devices which are responsible for the collection, storage and interpretation of the captured data. In addition, consumers, organizations, governments and companies themselves produce more and more data, for xample on social media. The amount of data is growing exponentially. People speak of Big Data when they work with one or more datasets that are too large to be maintained with regular database management systems.

يؤدي النمو الهائل للأنظمة القائمة على إنترنت الأشياء إلى زيادة مستمرة في أجهزة استشعار الإدخال المسؤولة عن جمع وتخزين وتفسير البيانات الملتقطة. بالإضافة إلى ذلك، ينتج المستهلكون والمنظمات والحكومات والشركات نفسها المزيد والمزيد من البيانات، على سبيل المثال على وسائل التواصل الاجتماعي. تتزايد كمية البيانات بشكل كبير. يتحدث الناس عن البيانات الضخمة عندما يعملون مع مجموعة بيانات واحدة أو أكثر كبيرة جدًا بحيث لا يمكن صيانتها باستخدام أنظمة إدارة قواعد البيانات العادية.

The pros of applying sensor data to the input devices is that the decisions of the IoT devices are subjected to information accessed from the evidences and not from any kind of irrelevant details and subjective experiences. This type of knowledge base makes the system cost effective with the enhanced streamlined processes, boosted product quality and better services. By combining data intelligently and by interpreting / translating, new insights are created that can be used for new services, applications and markets. This information can also be combined with data from various external sources, such as weather data or demographics.

تتمثل مزايا تطبيق بيانات الاستشعار على أجهزة الإدخال في أن قرارات أجهزة إنترنت الأشياء تخضع للمعلومات التي يتم الوصول إليها من الأدلة وليس من أي نوع من التفاصيل غير ذات الصلة والتجارب الذاتية. يجعل هذا النوع من قاعدة المعرفة النظام فعالاً من حيث التكلفة مع العمليات المبسطة المحسنة وجودة المنتج المعززة والخدمات الأفضل. من خلال الجمع بين البيانات بذكاء وعن طريق التفسير / الترجمة، يتم إنشاء رؤى جديدة يمكن استخدامها للخدمات والتطبيقات والأسواق الجديدة. يمكن أيضًا دمج هذه المعلومات مع البيانات من مصادر خارجية مختلفة، مثل بيانات الطقس أو التركيبة السكانية.

3.3.6 Graph/ Network Data

“**Graph data**” is the type of data which can be represented as **graph** with a special characteristic of comprising the mathematical graph theory into the mined information.

“Graph” in the network data generally represents a model in the statistical and mathematical domain with pair wise relationship in the constituent objects.

Graph or **network data** is, in short, data that focuses on the relationship or adjacency of **objects**.

The representation of the graph models includes the **nodes**, **edges**, and **relationships** between the stored data in the nodes.

Graph-based data is a natural way to represent **social networks**, and its structure allows you to calculate specific metrics such as the influence of a person and the shortest path between two people.

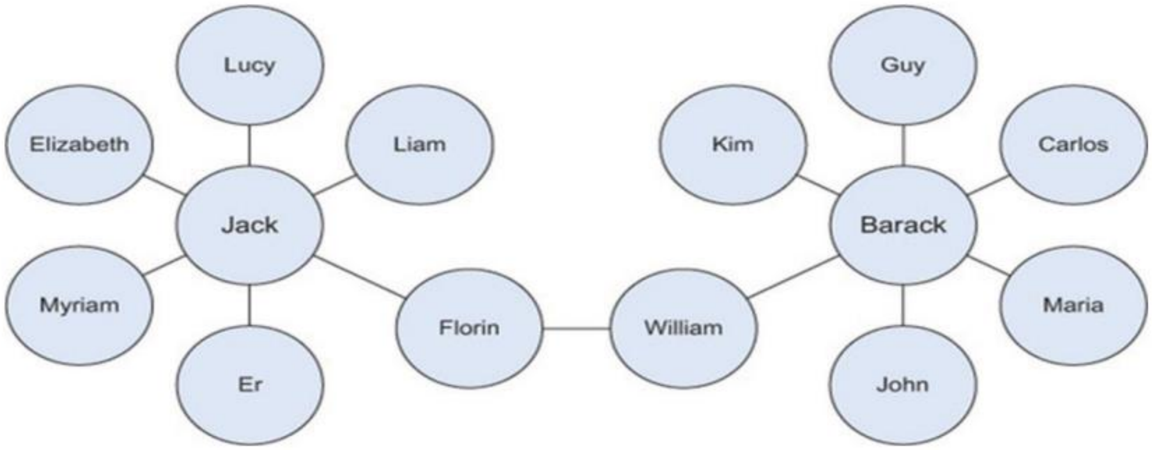


Fig. 3.6.3 Friends in social network are example of Graph Network

Examples of graph-based data can be found on many social media websites (figure 3.6.3).

For instance, on **LinkedIn** you can see who you know at which company.

Your follower list on **Twitter** is another example of graph-based data.

Graph databases are used to store graph-based data and are queried with specialized query languages such as **SPARQ**.

3.4 MULTIPLE PROBLEMS IN HANDLING LARGE DATASETS

The multiple problems in the large datasets are discussed by numerous data scientists with the need to understand the growing demand of the data and handling the mathematical as well as statistical operations for the manipulation of the data. In the last two years, over 90% of the world's data was created, and with 2.5 quintillion bytes of data generated daily, it is clear that the future is filled with more data, which can also mean more data problem in context to the following:

• Collecting, storing, sharing and securing data

• Creating and utilizing meaningful insights from their data.

Some common big data problems and the respective solutions have been discussed below:

1. Lack of Understanding

The lack of understanding of the mined data in the data science-based companies might lead to knocked down the performance in many areas. إن عدم فهم البيانات المستخرجة في الشركات القائمة على علم البيانات قد يؤدي إلى انخفاض الأداء في العديد من المجالات.

Many of the major areas for the data- based companies were:

depreciate the expenses of mining information, innovate new ideas for interpretation, new product launching, enhance performance and so on. Despite the benefits, companies have been slow to adopt data for a data centric approach.

وكانت العديد من المجالات الرئيسية للشركات القائمة على البيانات هي: خفض تكاليف استخراج المعلومات، وابتكار أفكار جديدة للتفسير، وإطلاق منتجات جديدة، وتحسين الأداء وما إلى ذلك. وعلى الرغم من الفوائد، كانت الشركات بطيئة في تبني البيانات لنهج يركز على البيانات.

Solution: Follow a top-down approach for the introduction and manipulation of the data science based on the procedures followed up. In case of lack of a data science professional, the consultancy services or an IT proficient with data science knowledge should be hired to get a better understanding.

اتبع نهجًا من أعلى إلى أسفل لتقديم علوم البيانات والتلاعب بها بناءً على الإجراءات المتبعة. في حالة عدم وجود متخصص في علوم البيانات، يجب الاستعانة بخدمات الاستشارات أو متخصص في تكنولوجيا المعلومات يتمتع بمعرفة علوم البيانات للحصول على فهم أفضل.

2. High Cost of Data Solutions

The companies have understood that buying and maintaining of necessary components make the system less cost effective. In addition to cost of the servers and the software-based storage, the high-end cost of the data science experts makes the system time consuming.

أدركت الشركات أن شراء وصيانة المكونات الضرورية يجعل النظام أقل فعالية من حيث التكلفة. بالإضافة إلى تكلفة الخوادم والتخزين القائم على البرامج، فإن التكلفة العالية لخبراء علوم البيانات تجعل النظام يستهلك وقتًا طويلاً.

Solution: The solution is to understand the need and use of the data with a collaborative method to find a goal, conduct a research or solution and implement the execution with a plan. الحل هو فهم الحاجة إلى البيانات واستخدامها بطريقة تعاونية للعثور على هدف وإجراء بحث أو حل و التنفيذ وفقًا لخطة.

3. Too Many Choices

Coined as the “paradox of choice” Schwartz explains how option overload can cause inaction on behalf of a buyer. In the world of data and data tools, the options are almost as widespread as the data itself, so it is understandably overwhelming when deciding the solution that‘s right for the business, especially when it will likely affect all departments and hopefully be a long-term strategy.

ويشرح شوارتز كيف يمكن أن يؤدي الإفراط في الخيارات إلى عدم اتخاذ أي إجراء من جانب المشتري، وهو ما يُعرَف باسم "مفارقة الاختيار". ففي عالم البيانات وأدوات البيانات، تنتشر الخيارات على نطاق واسع مثل البيانات نفسها، لذا فمن المفهوم أن يكون الأمر مرهقًا عند تحديد الحل المناسب للشركة، وخاصة عندما من المرجح أن يؤثر على جميع الأقسام، ونأمل أن يكون استراتيجية طويلة الأجل.

Solution: Like understanding data, a good solution is to leverage the experience of your in-house expert, perhaps a CTO. If that‘s not an option, hire a consultancy firm to assist in the decision-making process. Use the internet and forums to source valuable information and ask questions. وكما هو الحال مع فهم البيانات، فإن الحل الجيد هو الاستفادة من خبرة الخبير الداخلي، ربما أحد مسؤولي التكنولوجيا الرئيسيين. وإذا لم يكن هذا خيارًا متاحًا، فاستعن بشركة استشارية لمساعدتك في عملية اتخاذ القرار. استخدم الإنترنت والمنتديات للحصول على معلومات قيمة وطرح الأسئلة.

4. Complex Systems for Managing Data

The systems to understand the data management and finding a relevant solution for the manipulation of data is in itself a problem Due to the vast expanse of the different types of data with the IT teams creating their own data during the process of data handling results in increased complexity.

إن إيجاد أنظمة لفهم إدارة البيانات وإيجاد حل مناسب لمعالجة البيانات يعد في حد ذاته مشكلة نظرًا للامتداد الواسع للأنواع المختلفة من البيانات مع قيام فرق تكنولوجيا المعلومات بإنشاء بياناتها الخاصة أثناء عملية معالجة البيانات مما يؤدي إلى زيادة التعقيد.

Solution: Find a solution with a single command center, implement automation whenever possible, and ensure that it can be remotely accessed 24/7.

ابحث عن حل باستخدام مركز قيادة واحد، وقم بتنفيذ الأتمتة كلما أمكن ذلك، وتأكد من إمكانية الوصول إليه عن بعد على مدار الساعة طوال أيام الأسبوع.

5. Security Gaps

Another important aspect of the data science is the security of the data and the biasing of the large amount of data is always possible. In order to handle it the encryption and decryption must be performed with the data store with proper storage.

من الجوانب المهمة الأخرى لعلم البيانات هو أمان البيانات، حيث يكون من الممكن دائمًا تحيز كمية كبيرة من البيانات. وللتعامل مع هذا، يجب إجراء التشفير وفك التشفير باستخدام مخزن البيانات مع التخزين المناسب.

Solution: The data need to be handled with automated security updates of the data warehouse and automated backups.

6. Low Quality and Inaccurate Data

Having data is only useful when it‘s accurate. Low quality data not only serves no purpose, but it also uses unnecessary storage and can harm the ability to gather insights from clean data.

A few ways that data can be considered low quality is:

• Inconsistent formatting (which will take time to correct and can happen when the same elements are spelled differently like —"US" versus —"U.S."),

• Missing data (i.e. a first name or email address is missing from a database of contacts),

• Inaccurate data (i.e. it's just not the right information or the data has not be updated).

• Duplicate data (i.e. the data is being double counted)

• If data is not maintained or recorded properly, it's just like not having the data in the first place.

Solution: Begin by defining the necessary data you want to collect (again, align the information needed to the business goal). Cleanse data regularly and when it is collected from different sources, organize and normalize it before uploading it into any tool for analysis.

Once you have your data uniform and cleansed, you can segment it for better analysis.

7. Compliance Hurdles

When collecting information, security and government regulations come into play. With the somewhat recent introduction of the General Data Protection Regulation (GDPR), it‘s even more important to understand the necessary requirements for data collection and protection, as well as the implications of failing to adhere. Companies have to be compliant and careful in how they use data to segment customers for example deciding which customer to prioritize or focus on.

This means that the data must: be a representative sample of consumers, algorithms must prioritize fairness, there is an understanding of inherent bias in data, and Big Data outcomes have to be checked against traditionally applied statistical practices.

Solution: The only solution to adhere to compliance and regulation is to be informed and well-educated on the topic. There‘s no way around it other than learning because in this case, ignorance is most certainly not bliss as it carries both financial and reputational risk to your business. If you are unsure of any regulations or compliance you should consult expert legal and accounting firms specializing in those rules.

3.5 General Techniques for Handling Large Data

The multiple challenges have been discussed in the section above and the solutions for the defies are categorized based on the algorithms and out of memory errors. Never ending algorithms, out-of-memory errors, and speed issues are the most common challenges you face when working with large data. In this section, we'll investigate solutions to overcome or alleviate these problems.

The solutions can be divided into three categories: using the correct algorithms, choosing the right data structure, and using the right tools. (Figure 3.1.4)

There is no such relationship between the problems discussed in the section above and the solutions to be incorporated. There are multiple solutions to a given problem which can also handle the issue of memory and computational overhead.

This type of data science solutions can be generalized for challenges in the exploration of data.

For instance, the **compression** and **decompression** of the data set help resolve the memory issues but this also affects computation speed with a shift from the slow hard disk to the fast CPU. Contrary to RAM (random access memory), the hard disc will store everything even after the power goes down, but writing to disc costs more time than changing information in the fleeting RAM.

When constantly changing the information, RAM is thus preferable over the (more durable) hard disc.

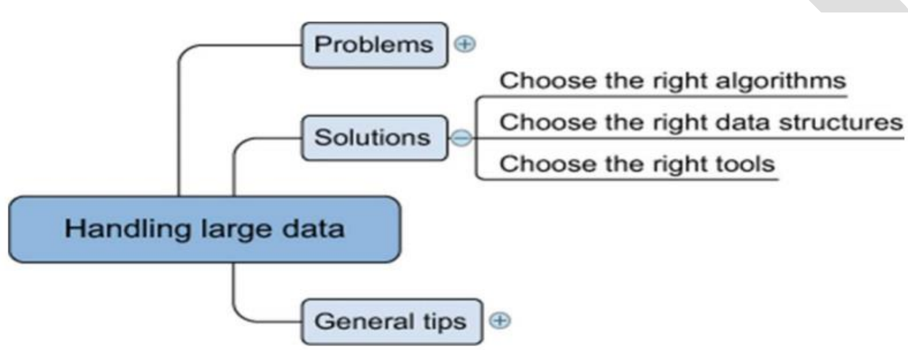


Fig. 3.1.4 Overview of solutions for handling large data sets

3.5.1 Choosing the Right Algorithm

The selection of an effective algorithm for the processing of the data can provide better results as compared to the enhanced hardware. In order to perform the predictive or selective analysis the **best-chosen algorithm** need not necessarily load the complete data into the memory or RAM, it rather supports the parallel computations with parallel or distributed databases.

In this section **three types of algorithms** have been discussed that can perform the computations parallelly reducing the computation or memory overhead:

online algorithms, **block algorithms**, and **MapReduce algorithms**, as shown in figure 3.1.5.

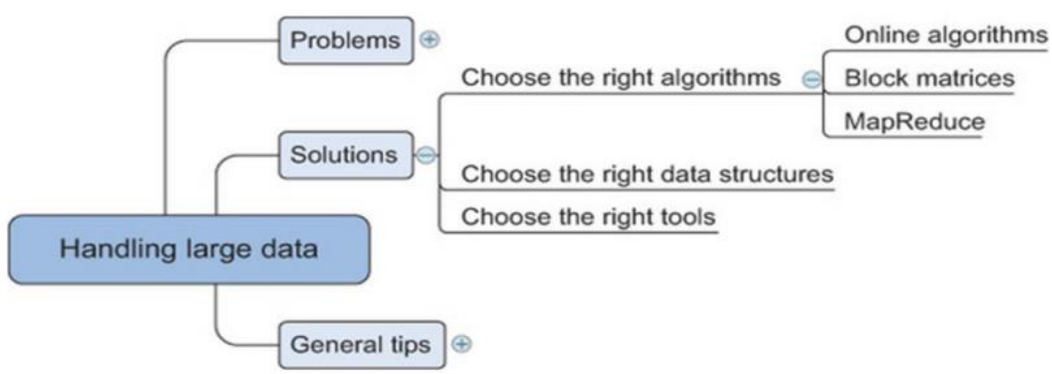


Fig. 3.1.5 The algorithms for handling the Big Data

Several, but not all, machine learning algorithms can be trained using one observation at a time instead of taking all the data into memory. The model can be trained based on the current parameters and the previous parameter values can be made to forget by the algorithm.

This technique of “use and forget “ helps to attain high memory effectiveness in the system.

This way as the new data values is acquired by the algorithm, the previous values are forgotten or overwritten but the effect can be witnessed or observed in the performance metrics of the proposed model. Most online algorithms can also handle mini-batches; this way, the data science exert or the developer can feed the batches of 10 to 1,000 observations at one single instance and then applying the sliding window protocol to access the data. This learning can be handled by multiple means discussed as follows:

- Full batch learning (also called statistical learning) —Feed the algorithm all the data at once.
- Mini-batch learning —Feed the algorithm a spoonful (100, 1000, ..., depending on what your hardware can handle) of observations at a time.
- Online learning —Feed the algorithm one observation at a time.

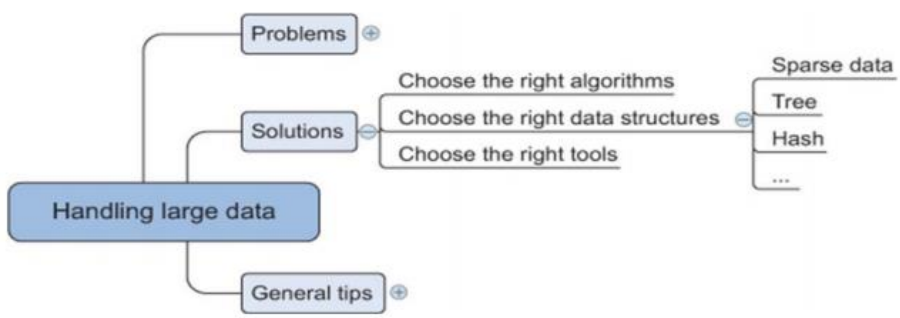


Fig. 3.1.6 Data structures applied in the data science

3.5.2 Choosing the Right Data Structure

The algorithms as discussed can enhance the performance and execution for the manipulation of the data in the warehouse. This process in the data science field actually leads to fragmentation in the raw data so that is the reason the structures for the storage of the data is equally important for the data scientist or a data science researcher. Data structures have different storage requirements, but also influence the performance of CRUD (create, read, update, and delete) and other operations on the data set.

• Sparse Data

Most of the IT professional and the data scientist understand the representation through a sparse matrix. Similarly, the sparse data illustration can be done by giving a minimal detail about the data as compared to the total number of entries.

As in figure 3.1.7, the conversion of the data from the text to binary or an image to binary can be expressed as the sparse data.

Imagine a set of 100,000 completely **unrelated Twitter tweets**. Most of them probably have fewer than 30 words, but together they might have hundreds or thousands of distinct words.

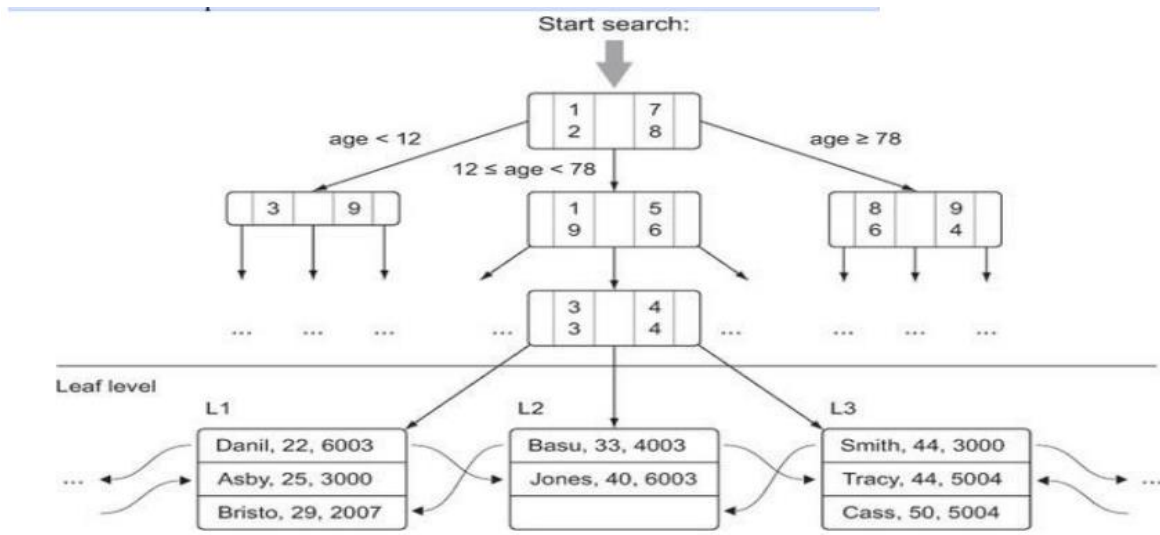
For this reason, the text documents are processed for the **stop words**, cut into **fragments** and stored as vectors instead of the binary information. The basic idea is that any **word present in the tweet is expressed as 1** and **not in tweet is expressed as 0** resulting in **sparse data** indeed. But the matrix generated would require equal memory as compared to any other matrix even though it has a little information.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

• Tree Structure

Trees is a special type of data structure that has faster retrieval of information in comparison to the table or sparse data.

In these data structures the root node is the first directory for accessing the information and the child nodes are the sub directories of the root node. The information can be accessed form the child or leaf nodes by either using pointers or indexing of the tree structure. The figure 3.8 helps the researcher to understand the process of information retrieval in the tree structure.



• Hash Tables

The process of the calculation of the key for each data entry and allotting the key to the relevant bucket is the important process of the hash table structure for th storage of data. The process of storage and handling in the hash table makes it more reliable source for the retrieval of information based on the key value. This way you can quickly retrieve the information by looking in the right bucket when you encounter the data. Dictionaries in Python are a hash table implementation, and they‘re a close relative of key-value stores. You’ll encounter them in the last example of this chapter when you build a recommender system within a database.

Hash tables are used extensively in databases as indices for fast information retrieval.

3.5.2Choosing the Right Tools

As discussed earlier in section 3.4.1 and 3.4.2 with the need to have a right algorithm and the right data structure, the requirement of a good tool is also important. The right tool can be a **Python library** or at least a tool that's controlled from Python, as shown figure 3.1.9.

Python has a number of libraries that can help you deal with large data. They range from smarter data structures over code optimizers to just-in-time compilers. The following is a list of libraries we like to use when confronted with large data:

Cython —The closer to the actual hardware of a computer, the more vital it is for the computer to know what types of data it has to process. For a computer, adding 1 + 1 is different from adding 1.00 + 1.00. The first example consists of integers and the second consists of floats, and these calculations are performed by different parts of the CPU.

Python needs no specifications of the types of the data but the compiler can itself interpret the values based on the integer or float. This is although a slow process and so a superset of python can be used for the solution which forces the user to define the data type before the execution and hence can be implemented much faster. See http://cython.org/ for more information on Cython.

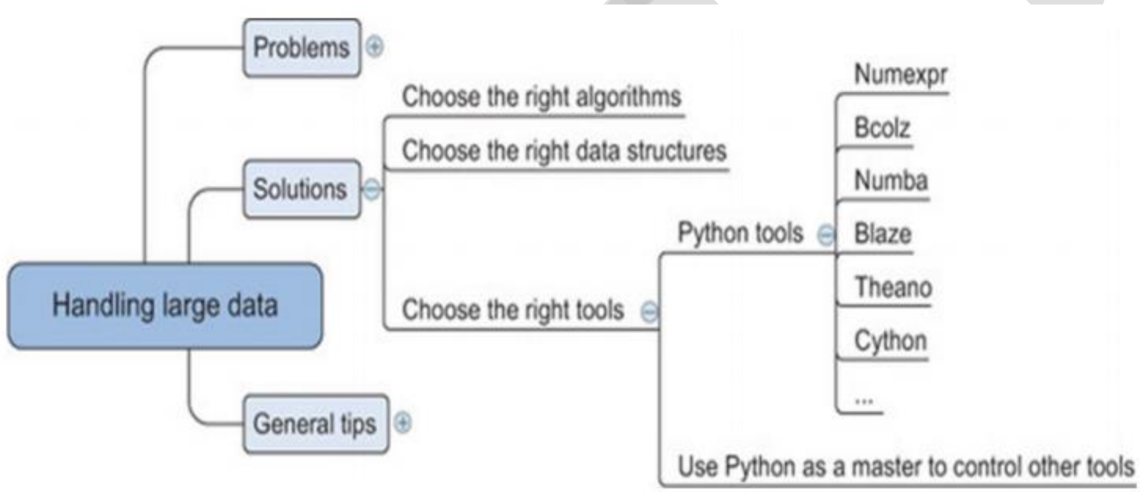


Fig. 3.1.7 Tools applied by data scientist for handing data

Numexpr —Numexpr is at the core of many of the big data packages, as is NumPy for in-memory packages. Numexpr is a numerical expression evaluator for NumPy but can be many times faster than the original NumPy. https://github.com/pydata/numexpr.

Numba —Numba is the package which compiles (just-in-time compiling) the code for the data manipulation and handling before actually executing it which makes it faster and less prone to errors. With the user gets a platform to develop a high level code wit the speed of the compilation as in C. http://numba.pydata.org/.

Bcolz —Bcolz helps you overcome the out-of-memory problem that can occur when using NumPy. It can store and work with arrays in an optimal compressed form. It not only slims down your data need but also uses Numexpr in the background to reduce the calculations needed when performing calculations with bcolz arrays. **http://bcolz.blosc.org/.**

Blaze — Blaze is ideal in case the data scientist use the power of a database backend but like the —Pythonic way‖ of working with data. Blaze will translate your Python code into SQL but can handle many more data stores than relational databases such as CSV, Spark, and others. Blaze delivers a unified way of working with many databases and data libraries.

http://blaze.readthedocs.org/en/latest/index.html.

Theano —Theano enables you to work directly with the graphical processing unit (GPU) and do symbolical simplifications whenever possible, and it comes with an excellent just-in-time compiler. On top of that it's a great library for dealing with an advanced but useful mathematical concept: tensors.

http://deeplearning.net/software/theano.

3.6 DISTRIBUTING DATA STORAGE AND PROCESSING WITH FRAMEWORKS

New big data technologies such as Hadoop and Spark make it much easier to work with and control a cluster of computers. Hadoop can scale up to thousands of computers, creating a cluster with petabytes of storage. This enables businesses to grasp the value of the massive amount of data available. ―Big data Analytics‖ is a phrase that was coined to refer to amounts of datasets that are so large, traditional data processing software simply can‘t manage them. For example, big data is used to pick out trends in economics, and those trends and patterns are used to predict what will happen in the future. These vast amounts of data require more robust computer software for processing, best handled by data processing frameworks. These are the top preferred data processing frameworks, suitable for meeting a variety of different needs of businesses.

3.6.1 Hadoop

This is an open-source batch processing framework that can be used for the distributed storage and processing of big data sets. Hadoop relies on computer clusters and modules that have been designed with the assumption that hardware will inevitably fail, and those failures should be automatically handled by the framework.

There are **four** main modules within Hadoop. Hadoop Common is where the libraries and utilities needed by other Hadoop modules reside. The Hadoop Distributed File System (HDFS) is the distributed file system that stores the data. Hadoop YARN (Yet Another Resource Negotiator) is the resource management platform that manages the computing resources in clusters, and handles the scheduling of users‘ applications. The

Hadoop MapReduce involves the implementation of the MapReduce programming model for large-scale data processing.

Hadoop operates by splitting files into large blocks of data and then distributing those datasets across the nodes in a cluster. It then transfers code into the nodes, for processing data in parallel. The idea of data locality, meaning that tasks are performed on the node that stores the data, allows the datasets to be processed more efficiently and more quickly.

Hadoop can be used within a traditional onsite datacenter, as well as through the cloud.

3.6.2 Apache Spark

Apache Spark is a batch processing framework that has the capability of stream processing, as well, making it a hybrid framework. Spark is most notably easy to use, and it's easy to write applications in Java, Scala, Python, and R. This open-source cluster computing framework is ideal for machine-learning, but does require a cluster manager and a distributed storage system. Spark can be run on a single machine, with one executor for every CPU core. It can be used as a standalone framework, and you can also use it in conjunction with Hadoop or Apache Mesos, making it suitable for just about any business.

Spark relies on a data structure known as the Resilient Distributed Dataset (RDD).

This is a read-only multiset of data items that is distributed over the entire cluster of machines. RDDs operate as the working set for distributed programs, offering a restricted form of distributed shared memory. Spark is capable of accessing data sources like HDFS, Cassandra, HBase, and S3, for distributed storage. It also supports a pseudo distributed local mode that can be used for development or testing.

The foundation of Spark is Spark Core, which relies on the RDD-oriented functional style of programming to dispatch tasks, schedule, and handle basic I/O functionalities. Two restricted forms of shared variables are used: broadcast variables, which reference read-only data that has to be available for all the nodes, and accumulators, which can be used to program reductions. Other elements included in Spark Core are:

Spark SQL, which provides domain-specific language used to manipulate Data Frames.

Spark Streaming, which uses data in mini-batches for RDD transformations, allowing the same set of application code that is created for batch analytics to also be used for streaming analytics. Spark MLlib, a machine-learning library that makes the large-scale machine learning pipelines simpler. GraphX, which is the distributed graph processing framework at the top of Apache Spark.

3.6.3 Apache Storm

This is another open-source framework, but one that provides distributed, real-time stream processing. Storm is mostly written in Clojure, and can be used with any programming language. The application is designed as a topology, with the shape of a Directed Acyclic Graph (DAG). Spouts and bolts act as the vertices of the graph. The idea behind Storm is to define small, discrete operations, and then compose those operations into a topology, which acts as a pipeline to transform data.

3.6.4 Samza

Samza is another open-source framework that offers near a real-time, asynchronous framework for distributed stream processing. More specifically, Samza handles immutable streams, meaning transformations create new streams that will be consumed by other components without any effect on the initial stream. This framework works in conjunction with other frameworks, using Apache Kafka for messaging and Hadoop YARN for fault tolerance, security, and management of resources.

3.6.5 Flink

Flink is a hybrid framework, open-source, and stream processes, but can also manage batch tasks. It uses a high-throughput, low-latency streaming engine that is written in Java and Scala, and the runtime system that is pipelined allows for the execution of both batch and stream processing programs. The runtime also supports the execution of iterative algorithms natively. Flink's applications are all fault-tolerant and can support exactly-once semantics. Programs can be written in Java, Scala, Python, and SQL, and Flink offers support for event-time processing and state management.

3.8 QUESTIONS

1. Discuss the various representation techniques of different data types.

2.What are the issues that are faced in handling large data?

3. Explain briefly the techniques to handle large data.

4. What are the popular frameworks for big data storage?